

CravingProbe: A System Combining Virtual Reality and Biofeedback Technologies to Assist Drug Psychotherapy

Chieh-Jui Ho*
National Tsing Hua University
jerryjp3914@gapp.nthu.edu.tw

Chi-Ting Hou*
National Tsing Hua University
dafdafdaf0411@gapp.nthu.edu.tw

Min-Wei Hung*
National Tsing Hua University
mwhung@mx.nthu.edu.tw

Chien Wen (Tina) Yuan
National Taiwan Normal University
tinayuan@ntnu.edu.tw

Nanyi Bi
National Taiwan University
nanyi.bi@gmail.com

Ming-Chyi Huang
Taipei City Hospital Songde Branch
mch@tpech.gov.tw

Chuang-Wen You
National Tsing Hua University
cwyou@mx.nthu.edu.tw

ABSTRACT

Virtual Reality has demonstrated its potential in drug psychotherapy. However, there is little research on the design space of VR experience combined with biofeedback in clinical settings for drug addiction that is aligned with treatment goals. We designed a prototype system, called *CravingProbe*, combining VR and biofeedback technologies to assist drug psychotherapy, and recruited three psychotherapists for a pilot user study to share their feedback from therapeutic perspectives. Results show that participants agreed that the system illustrates most of the typical real-life craving episodes and useful to discuss potential cravings experienced by patients. And, participants expressed willingness to and felt capable of using this system during drug psychotherapy.

CCS CONCEPTS

• Human-centered computing → Ubiquitous and mobile computing systems and tools.

KEYWORDS

Virtual Reality; Biofeedback; Psychotherapy; Drug Addiction

ACM Reference Format:

Chieh-Jui Ho, Chi-Ting Hou, Min-Wei Hung, Chien Wen (Tina) Yuan, Nanyi Bi, Ming-Chyi Huang, and Chuang-Wen You. 2021. CravingProbe: A System Combining Virtual Reality and Biofeedback Technologies to Assist Drug Psychotherapy. In *Adjunct Proceedings of the 2021 ACM International Joint Conference on Pervasive and Ubiquitous Computing and Proceedings of the 2021 ACM International Symposium on Wearable Computers (UbiComp-ISWC '21 Adjunct)*, September 21–26, 2021, Virtual, USA. ACM, New York, NY, USA, 5 pages. <https://doi.org/10.1145/3460418.3480159>

*These authors contributed equally to this research.

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than the author(s) must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.
UbiComp-ISWC '21 Adjunct, September 21–26, 2021, Virtual, USA
© 2021 Copyright held by the owner/author(s). Publication rights licensed to ACM.
ACM ISBN 978-1-4503-8461-2/21/09...\$15.00
<https://doi.org/10.1145/3460418.3480159>

1 INTRODUCTION

Drug addiction is a complex disorder manifested by compulsive drug use, which would present various symptoms, such as emotional irritability or depression. Cognitive Behavioral Therapy (CBT) [23] is one of the therapeutic approaches to effectively assist drug-addicted patients in rehabilitation. In CBT for drug addiction, therapists focus on guiding patients to recollect the moments when craving arise, and review the triggering factors, bodily responses, and mental experiences under the circumstances. Only when patients have the capacity to identify triggering factors and improve their craving awareness can they practice self-regulation and coping skills, thereby blocking the inertial relapse process at the right time.

Virtual Reality (VR) has been introduced in CBT as a tool for assessment and intervention, and has demonstrated its potential in VR exposure therapy (VRET) [28]. However, in treating drug addiction, the focus is mostly on the effectiveness and validity to induce craving [13]. Moreover, integration of VR and biofeedback in HCI community has been explored specifically in the application domain of relaxation and gaming experience. To the best of our knowledge, there is little research on the qualitative insights on the design space of VR experience combined with biofeedback in clinical settings for drug addiction that is aligned with treatment goals. Prior to this study, an interview study was conducted with psychotherapists for drug addiction treatment in regards to the issues concerning the application of VR and biofeedback in drug psychotherapy [19]. The study then proposed a preliminary system design that provides a clinical-friendly VR experience that incorporates timely biofeedback in an effort to facilitate treatment progress of drug addiction [19].

Built on prior studies, we designed a prototype system, called *CravingProbe*, combining virtual reality and biofeedback technologies to assist drug psychotherapy, and recruited three psychotherapists for a pilot study to share their feedback from a therapeutic perspective. Results show that participants agreed that the system illustrates most of the typical real-life craving episodes and is useful to discuss potential cravings experienced by patients. And, participants expressed willingness to and felt capable of using this system during drug psychotherapy.

2 RELATED WORK

We summarize studies on how to apply (1) virtual reality, (2) biofeedback technologies, and (3) VR and biofeedback combined to assist psychotherapists in drug addiction treatment, described as follows.

2.1 VR Applications in Psychotherapy

The adaptation of VR in CBT has been the interest of the psychotherapy field [14, 21, 22, 26]. The application of VR in psychotherapy may improve the quality of cues by making vivid visual reconstructions, providing a first-person immersive experience in a virtual scene, and adding details that are difficult to reproduce in reality. With the assistance of VR, therapists can use lifelike stimuli to assess craving, and improve the effectiveness of exposure techniques (Virtual Reality Exposure Therapy, VRET). Despite many advantages of VR applications, there are still limitations to the therapeutic use of VR. For instance, pre-designed content without customization that is far from user's real-life experience may result in difficulties for users to immerse themselves in the virtual environment and transfer improvements in treatment into daily life [18]. Besides, exposing patients to virtual cues that arouse intense craving may put patients at risk despite original purpose, which demands careful consideration in design [19]. Therefore, how to integrate VR technology into the clinical practice of drug addiction treatment and align with the treatment goal still invites further exploration from a user-centered perspective.

2.2 Biofeedback Technologies in Psychotherapy

Biofeedback therapy [17] is the use of monitoring devices to visualize physiological responses, which may only be perceived by sensing technology, allowing not only the patient to understand the interaction between body and mind but also therapists to gain an objective understanding of the patient's state. Therapists can then use collected data to guide the patient to improve mind-body awareness and learn to self-regulate along with visual representations [15, 16, 25]. To ensure the validity of craving assessment, researchers and practitioners often use patient's self-report along with multiple physiological indicators [12, 30, 32, 34, 35], commonly including heart rate variability (HRV), skin conductance (SC), skin temperature (TEMP), or electrocardiogram (ECG). When assessing craving with biofeedback in drug addiction psychotherapy, therapists help attach biofeedback sensors to patients and record the baseline of their physiological state. Afterwards, therapists instruct patients to recall the situation in which their drug craving occurred and collect patients' bodily responses as indicators of craving. Once the patient's state is aroused, it is essential for the therapists to make sure that the aroused state of patients goes back to baseline by providing self-regulation techniques [10, 31].

2.3 Combining VR and Biofeedback for Drug Psychotherapy

In recent years, the combination of VR and biofeedback has become popular in mindfulness and relaxation exercises [24, 27, 29]. In addition, VR psychotherapy often incorporates biofeedback to monitor patient's physiological responses to a target virtual situation, so

that he/she can review the readings afterwards to pinpoint and reflect on specific, meaningful time points in the VR experience [22]. Repetto et al. [26] presented a clinical protocol for treating anxiety based on a biofeedback enhanced-VR system, and other VR-based approaches combined with biofeedback are explored in [11, 24, 33] to address issues around depression or anxiety.

However, so far little has been discussed for drug addiction as well as the design approach that is clinical-friendly. In light of this need, this study designed a prototype system, called *Craving-Probe*, based on the insights derived from a prior needs assessment study with psychotherapists [19] to further probe its feasibility. This prototype system integrated craving-induced virtual reality, real-time biofeedback, and an easy-to-use user interface to assist psychotherapists in drug psychotherapy.

3 SYSTEM DESIGN

3.1 System Architecture

The *CravingProbe* system contains four components: (1) a Virtual Reality (VR) headset [9], (2) a biofeedback device [1], (3) a database service [2], and (4) a Unity user interface. Patients first wear VR headsets and attach biofeedback sensors to their fingers. The VR headset is connected with a data server through a cable to stream VR contents for psychotherapists to review. The biofeedback devices collect patients' physiological data simultaneously, and then wirelessly upload the collected data to the database service running locally on the data server. Once the data is received, the database service will store the data and automatically update it onto the Unity user interface. The unity user interface will provide both real-time data updating and offline data querying functions to visualize patients' physiological responses in a comprehensive manner. In the following, we describe how these components work.

Figure 1 shows the Unity user interface, which is a custom editor window prototyped in Unity engine [7] consisting of four displaying areas displaying (1) time-series biofeedback data, (2) biofeedback snapshot, (3) self-report modes, and (4) VR content. According to the literature [12, 30, 32, 34, 35], heart rate variability (HRV), skin conductance (SC), and skin temperature (TEMP) typically change significantly while patients experience craving. So, the area of time-series biofeedback data plots linear graphs of the three selected types of physiological data with respect to the time. When psychotherapists use one of the areas to zoom in on the data of interest, the user interface automatically extracts the corresponding segment of data and updates the visualization in the other two.

When patients are watching the VR content, psychotherapists can easily spot the fluctuations of these readings from the visualization of the physiological data over time. Psychotherapists can directly check patients' current status from the area showing patients' biofeedback snapshots. Psychotherapists can observe patients' responses and actions to manually label a fragment of the data as craving- or ignored-type of events by tapping a key on the keyboard. After finishing the VR experience, psychotherapists can jointly use the time-series biofeedback data and the corresponding VR contents to pinpoint the potential craving events and discuss with patients to improve their awareness and understanding of craving.

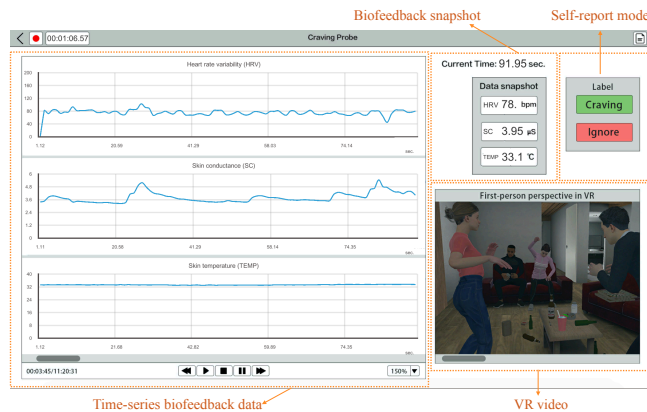


Figure 1: Unity User interface.

3.2 Interaction Stage in Psychotherapy Sessions

As illustrated in Figure 3, we proposed a five-stage protocol to utilize the *CravingProbe* system in chronological order (from top to bottom), which are described as follows.

3.2.1 Preparing stage. The purpose of this stage is to prepare the participant on what to expect with the system. Once psychotherapists decided to use the system, they first explain the goal of the VR-based therapy to participants, give instructions on what to do during the VR experience, and then assist the patients in putting up the VR headsets and biofeedback sensors. Before participants entering the VR experience, psychotherapists need to make sure that the data can be smoothly collected and visualized through the Unity UI.

3.2.2 Entering stage. The purpose of the entering stage is to guarantee participants a smooth transition from the physical to a virtual world and collect baseline values pertaining to HRV, SC, and TEMP. Upon entering the VR world, participants are first situated in a virtual copy of the consulting room to reduce their perceived spatial differences [20]. Participants are free to explore the space to slowly get used to the virtual world in a safe and familiar environment (in Figure 2(a)) to reduce the potential motion sickness [3]. Then, participants will watch a VR tutorial on how to interact with the system through controllers. Once the participants feel comfortable within VR, they can click a button to enter the normal home environment (in Figure 2(b)), which will contain triggering cues when participants revisit the inducing stage. After entering the normal home environment, participants follow the instruction in VR to relax their bodies to return to their baseline (the state in which participants remain calm). Then, the participants will be instructed to stand still for about a minute to collect baseline physiological signals (i.e., signals of HRV, SC, and TEMP). For psychotherapists, the baseline information serves as a reference for identifying notable changes in biofeedback readings.

3.2.3 Inducing stage. Leveraging the previous studies [19], we incorporated the four scenarios and modified the VR environment by placing more items (e.g., lighters potentially used for burning drugs) that might induce craving, and dimmed the lighting to create

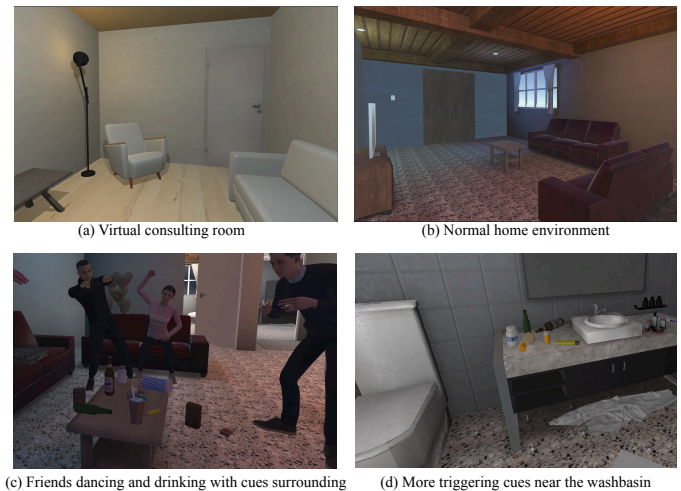


Figure 2: Example VR scenes created based on free assets from the Unity Asset Store [6], Sketchfab [5], and Mixamo [4].

a party atmosphere. The craving-inducing scenes start with the viewer standing in a bedroom, where the viewer can see a drug-free calendar (cue #1) suggesting abstinence of drug for nearly a month. After receiving a phone invitation from a friend 'Joe' (cue #2), the scene transitions to Joe's place, where a party is being held and the viewer would feel hyper when seeing a bunch of people dancing and drinking alcohol under a gloomy light (cue #3 in Figure 2(c)). In the living room, the viewer can spot food, bottles, and other triggering items (including a drinking cup with a straw or lighters) (cue #4). The viewer then walks to the toilet to view a private text message sent from a drug dealer (cue #5). Afterward, the viewer will wander back and enter a bedroom where 'Joe' is there standing still and staring at his hands, which illustrates the fixation behavior after drug use. Finally, the story ends with the viewer getting an after-party invitation from another friend, suggesting an upcoming drug supply (cue #6).

When participants are watching the VR content, psychotherapists monitor the real-time biofeedback data updated from the database service. If psychotherapists identify any abrupt change in readings similar to craving reactions [12, 30, 32, 34, 35], they can manually label the period as potential craving events using the Unity UI. Also, psychotherapists can observe participants' reactions to see if any confounding events (e.g., moving the biofeedback device) happens, which can be labeled as ignored. All the stored biofeedback data and the labeled events can be used in the reflecting stage.

3.2.4 Exiting stage. Given that exiting VR may also present similarities with entering VR [20], the participant will return to the virtual consulting room and then back to reality to reduce the perceived spatial differences. Psychotherapists will assist patients to remove the VR headset and biofeedback devices. If the participant reports feeling sick or remaining aroused by cues, psychotherapists will guide the participant to take a rest to relieve from potential exiting symptoms and/or to calm down with self-regulation exercises.

Stages	Therapist	Patient
Preparing	• Provide instructions & guidances • Assist in wearing biofeedback & VR devices	• Learn how to interact with the system • Wearing biofeedback & VR devices
Entering	• Observing patients' responses • Monitoring data collection on patient's baseline <i>*No direct interaction with patient</i>	• Transferring from physical to virtual environments • Follow instructions in VR to relax his/her body back to baseline <i>*No direct interaction with therapist</i>
Craving-Inducing	• Observing patients' responses • Label potential craving response via the Unity UI <i>*No direct interaction with patient</i>	• Watch VR videos and follow the predefined storyline • Exposed to virtual cues that induce craving <i>*No direct interaction with therapist</i>
Exiting	• Observing patients' responses • Assisting in transferring back to physical environment • Removing biofeedback & VR devices	• Removing biofeedback & VR devices • Transferring back to physical environment
Reflecting	• Review potential craving events with patients via Unity UI • Discuss craving cues and bodily response with patients	• Reflecting on potential craving events with the therapist • Discuss craving cues and bodily response with therapists

Figure 3: Interaction stages

3.2.5 Reflecting stage. The goal of this final stage is to review the potential craving events triggered by VR contents and bring in timely or suitable interventions to reinforce awareness and correct attributions of cravings. Psychotherapists first ask the participants to reflect if they felt craving for drugs and share their thoughts when watching scenes in the VR. Psychotherapists will clarify with the participants to pinpoint the cues that triggered their cravings. Then, psychotherapists will guide them to identify their body reactions based on the collected data and labels. By reviewing the data together, psychotherapists can better zoom in on the periods that contain cues participants may have overlooked. With the biofeedback 'evidence', psychotherapists can provide more interventions to reinforce participants' awareness, especially when they deny having any craving triggered by those cues.

4 PILOT USER STUDY

We conducted a user study involving psychotherapists, which was approved by the Institutional Review Board of Taipei City Hospital (IRB No.: TCHIRB-11001008).

4.1 Participants

From Taipei City Hospital, we recruited 3 psychotherapists (2 males and one female), aged from 29 to 36 years old ($M = 32$ years old; $SD = 3.61$), with 2.5 ~ 4 years of experience in conducting drug psychotherapy. All participants have a passing knowledge about virtual reality (VR) and biofeedback and have heard of applications of these technologies in psychotherapy [14, 21, 22, 26].

4.2 Experimental Environment

Due to the COVID-19 outbreak, we avoided direct contact with hospital personnel (as they have a high risk to be infected) or shared use of VR headsets and biofeedback devices. Instead, we interviewed participated psychotherapists online through a video conferencing tool. For participants to experience the VR scenes, we put our VR video on the Unity WebGL Platform [8] so that participants can watch it on their browsers using the Unity WebGL Player.

4.3 Procedure

The study procedure consisted of three phases. First, a pre-study orientation is conducted to explain the goal of this study and advise them on the operations to view the VR video of the CravingProbe system with their browsers. In the online experience and demo phase, participants load the pre-designed VR video by Unity WebGL Player running on their browsers and also share their browser screen over the video conferencing tool. After participants complete loading the VR video, they are encouraged to think aloud while exploring the VR scenes described in Section 3.2.3. After participants finish the VR video, a researcher will continue to explain the architecture of *CravingProbe* and the interaction stages when using this system in drug psychotherapy. Once participants understand how *CravingProbe* works, a researcher will remotely demo the system and show participants how the data will be updated through the Unity user interface. Finally, a post-study questionnaire (as described later) gathers the user feedback, which is ranked in a 7-point Likert scale (1 ("Strongly Disagree") ~ 7 ("Strongly Agree")), after seeing the demo of the system.

4.4 Preliminary Results

The four questions assessed how well the craving-inducing virtual reality scenes illustrate real-life craving episodes described by patients (*Qa*), if the system is useful to discuss potential craving experienced by patients (*Qb*), the willingness to incorporate this system in drug psychotherapy sessions (*Qc*), and if the participants feel they are capable of operating the system based on their clinical experience and professional skills (*Qd*). Overall, participants agreed that the virtual reality scenes illustrate most of the typical real-life craving episodes according to the average rating (5) towards *Qa*. However, participants also expressed that the intrinsic factors, such as stress responses, are also important and common in patients' real-life experiences and could be included in our future directions. Besides, participants felt positive about the usefulness of the system to discuss potential cravings experienced by patients (according to the average rating (5.67) towards *Qb*). They commented that, with this system, they could integrate their observation and the data to improve patients' understandings of triggering factors and bodily sensations. Also, participants expressed willingness to use this system (according to the average rating (6.33) *Qc*) and felt capable of operating this system (according to the average rating (6) towards *Qd*) during drug psychotherapy.

5 CONCLUSION & FUTURE WORK

We will continue to improve the system design and conduct a pilot field study to validate the feasibility to incorporate this system into real practices of psychotherapy sessions.

ACKNOWLEDGMENTS

This research was supported by the Ministry of Science and Technology of Taiwan (MOST 109-2221-E-007-114, 110-2221-E-007-098, and 109-2221-E-003-010-MY3).

REFERENCES

- [1] eVu TPS – The Wireless Triple-Physiology Sensor for Smartphones and Tablets. <https://evutps.com/en/>

- [2] Firebase. <https://firebase.google.com/>
- [3] How to Reduce Motion Sickness in Virtual Reality. <https://www.wired.com/story/how-to-reduce-motion-sickness-virtual-reality/>
- [4] Mixamo. <https://www.mixamo.com>
- [5] Sketchfab - The best 3D viewer on the web. <https://sketchfab.com>
- [6] Unity Asset Store. <https://assetstore.unity.com/>
- [7] Unity Real-Time Development Platform | 3D, 2D VR & AR Engine. <https://unity.com/>
- [8] Unity WebGL Platform. <https://docs.unity3d.com/Manual/webgl.html>
- [9] VIVE Cosmos Elite Overview | VIVE United States. <https://www.vive.com/us/product/vive-cosmos-elite/overview/>
- [10] Nour Alayan, Lucille Eller, Marsha E. Bates, and Dennis P. Carmody. 2018. Current Evidence on Heart Rate Variability Biofeedback as a Complementary Anticraving Intervention. *The Journal of Alternative and Complementary Medicine* 24, 11 (2018), 1039–1050. <https://doi.org/10.1089/acm.2018.0019> PMID: 29782180.
- [11] Hanshu Cai, Ziyang Wang, Yanhao Zhang, Yunfei Chen, and Bin Hu. 2017. A virtual-reality based neurofeedback game framework for depression rehabilitation using pervasive three-electrode EEG collector. In *Proceedings of the 12th Chinese Conference on Computer Supported Cooperative Work and Social Computing*, 173–176.
- [12] Stephanie Carreiro, Keerthi Kumar Chintha, Sloke Shrestha, Brittany Chapman, David Smelson, and Premananda Indic. 2020. Wearable sensor-based detection of stress and craving in patients during treatment for substance use disorder: A mixed methods pilot study. *Drug and alcohol dependence* 209 (2020), 107929.
- [13] Christopher Culbertson, Sam Nicolas, Itay Zaharovits, Edythe D. London, Richard De La Garza, Arthur L. Brody, and Thomas F. Newton. 2010. Methamphetamine Craving Induced in An Online Virtual Reality Environment. *Pharmacology Biochemistry and Behavior* 96, 4 (2010), 454–460. <https://doi.org/10.1016/j.pbb.2010.07.005>
- [14] Karlo Miguel R De Asis, Ethan Jeriko P Guillem, Francesca Angela M Reyes, and Mary Jane C Samonte. 2020. Serenity: A Stress-Relieving Virtual Reality Application based on Philippine Environmental Variables. In *Proceedings of the 2020 The 6th International Conference on Frontiers of Educational Technologies*, 155–159.
- [15] Jiang Du, Chenglu Fan, Haifeng Jiang, Haiming Sun, Xu Li, and Min Zhao. 2014. Biofeedback Combined With Cue-exposure as A Treatment for Heroin Addicts. *Physiology & Behavior* 130 (2014), 34–39. <https://doi.org/10.1016/j.physbeh.2014.02.055>
- [16] David Eddie, Evgeny Vaschillo, Bronya Vaschillo, and Paul Lehrer. 2015. Heart Rate Variability Biofeedback: Theoretical basis, delivery, and its potential for the treatment of substance use disorders. *Addiction Research & Theory* 23, 4 (2015), 266–272. <https://doi.org/10.3109/16066359.2015.1011625> arXiv:https://doi.org/10.3109/16066359.2015.1011625
- [17] Dana L. Frank, Lamees Khorshid, Jerome F. Kiffer, Christine S. Moravec, and Michael G McKee. 2010. Biofeedback in Medicine: who, when, why and how? *Mental health in family medicine* 7, 2 (2010), 85.
- [18] P. Fuchs and G. Moreau. 2003. *Le traité de la réalité virtuelle*. Number v. 1 in *Le traité de la réalité virtuelle*. Presses de l'Ecole des Mines. <https://books.google.com.tw/books?id=CrM9NAAACAAJ>
- [19] Min-Wei Hung, Chi-Ting Hou, Chieh-Jui Ho, Chien Wen (Tina) Yuan, Nanyi Bi, Shu-Huei Chen, Ming-Chyi Huang, and Chuang-Wen You. 2021. Exploring the Opportunities and Challenges of Enabling Clinical-Friendly Drug Psychotherapy with Virtual Reality and Biofeedback Technologies. In *Extended Abstracts of the 2021 CHI Conference on Human Factors in Computing Systems* (Yokohama, Japan) (CHI EA '21). Association for Computing Machinery, New York, NY, USA, Article 278, 7 pages. <https://doi.org/10.1145/3411763.3451585>
- [20] Jarrod Knibbe, Jonas Schjerlund, Mathias Petraeus, and Kasper Hornbæk. 2018. The Dream is Collapsing: The Experience of Exiting VR. In *Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems* (Montreal QC, Canada) (CHI '18). Association for Computing Machinery, New York, NY, USA, 1–13. <https://doi.org/10.1145/3173574.3174057>
- [21] Kaylee Payne Kruzan, Janis Whitlock, Natalya N Bazarova, Katherine D Miller, Julia Chapman, and Andrea Stevenson Won. 2020. Supporting Self-Injury Recovery: The Potential for Virtual Reality Intervention. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems*, 1–14.
- [22] Alice J. Lin, Fuhua (Frank) Cheng, and Charles B. Chen. 2020. Use of Virtual Reality Games in People with Depression and Anxiety. In *Proceedings of the 5th International Conference on Multimedia and Image Processing* (Nanjing, China) (ICMIP '20). Association for Computing Machinery, New York, NY, USA, 169–174. <https://doi.org/10.1145/3381271.3381299>
- [23] R. Kathryn McHugh, Bridget A. Hearon, and Michael W Otto. 2010. Cognitive Behavioral Therapy for Substance Use Disorders. *Psychiatric Clinics* 33, 3 (2010), 511–525.
- [24] Seulki Min, Jung-geun Moon, Chul-Hyun Cho, and Gerard J. Kim. 2020. Effects of Immersive Virtual Reality Content Type to Mindfulness and Physiological Parameters. In *26th ACM Symposium on Virtual Reality Software and Technology* (Virtual Event, Canada) (VRST '20). Association for Computing Machinery, New York, NY, USA, Article 24, 9 pages. <https://doi.org/10.1145/3385956.3418942>
- [25] Ana Isabel Penzlin, Timo Siepmann, Ben Min-Woo Illigens, Kerstin Weidner, and Martin Siepmann. 2015. Heart Rate Variability Biofeedback in Patients with Alcohol Dependence: A Randomized Controlled Study. *Neuropsychiatric disease and treatment* 11 (09 Oct 2015), 2619–2627. <https://doi.org/10.2147/NDT.S84798.26557753>[pmid].
- [26] Claudia Repetto, Andrea Gaggioli, Federica Pallavicini, Pietro Cipresso, Simona Raspelli, and Giuseppe Riva. 2013. Virtual Reality and Mobile Phones in the Treatment of Generalized Anxiety Disorders: A Phase-2 Clinical Trial. *Personal and Ubiquitous Computing* 17, 2 (2013), 253–260.
- [27] Christoph Rockstroh, Johannes Blum, and Anja S. Göritz. 2020. A mobile VR-based respiratory biofeedback game to foster diaphragmatic breathing. *Virtual Reality* (05 Oct 2020). <https://doi.org/10.1007/s10055-020-00471-5>
- [28] Barbara Olasov Rothbaum, Larry Hodges, and Rob Kooper. 1997. Virtual Reality Exposure Therapy. *Journal of Psychotherapy Practice & Research* (1997).
- [29] Eunbi Seol, Seulki Min, Sungho Seo, Seoyeon Jung, Youngil Lee, Jaedong Lee, Gerard Kim, Chungyeon Cho, Seungmoo Lee, Chul-Hyun Cho, Seungmoon Choi, and Dooyoung Jung. 2017. "Drop the Beat": Virtual Reality Based Mindfulness and Cognitive Behavioral Therapy for Panic Disorder — a Pilot Study. In *Proceedings of the 23rd ACM Symposium on Virtual Reality Software and Technology* (Gothenburg, Sweden) (VRST '17). Association for Computing Machinery, New York, NY, USA, Article 57, 3 pages. <https://doi.org/10.1145/3139131.3141199>
- [30] Haoye Tan, Tianzhen Chen, Jiang Du, Runji Li, Haifeng Jiang, Cheng-Long Deng, Weidong Song, Ding Xu, and Min Zhao. 2019. Drug-related Virtual Reality Cue Reactivity is Associated with Gamma Activity in Reward and Executive Control Circuit in Methamphetamine Use Disorders. *Archives of medical research* 50, 8 (2019), 509–517.
- [31] Bryan K. Tolliver, Aimee L. McRae-Clark, Michael Saladin, Kimber L. Price, Annie N. Simpson, Stacia M. DeSantis, Nathaniel L. Baker, and Kathleen T. Brady. 2010. Determinants of Cue-Elicited Craving and Physiologic Reactivity in Methamphetamine-Dependent Subjects in the Laboratory. *The American Journal of Drug and Alcohol Abuse* 36, 2 (2010), 106–113. <https://doi.org/10.3109/00952991003686402> arXiv:https://doi.org/10.3109/00952991003686402 PMID: 20337507.
- [32] Meng-Chang Tsai, Chia-Ru Chung, Chun-Chuan Chen, Jyun-Yu Chen, Shih-Ching Yeh, Chun-Han Lin, Ying-Ju Chen, Ming-Che Tsai, Ya-Ling Wang, Chia-Ju Lin, and Eric Hsiao-Kuang Wu. 2021. An Intelligent Virtual-Reality System With Multi-Model Sensing for Cue-Elicited Craving in Patients With Methamphetamine Use Disorder. *IEEE Transactions on Biomedical Engineering* 68, 7 (2021), 2270–2280. <https://doi.org/10.1109/TBME.2021.3058805>
- [33] Marieke Van Rooij, Adam Lobel, Owen Harris, Niki Smit, and Isabela Granic. 2016. DEEP: A Biofeedback Virtual Reality Game for Children At-risk for Anxiety. In *Proceedings of the 2016 CHI Conference Extended Abstracts on Human Factors in Computing Systems*, 1989–1997.
- [34] Yong-Guang Wang, Zhi-Hua Shen, and Xuan-Chen Wu. 2018. Detection of patients with methamphetamine dependence with cue-elicited heart rate variability in a virtual social environment. *Psychiatry research* 270 (2018), 382–388.
- [35] Min Zhao, Chenglu Fan, Jiang Du, Haifeng Jiang, Hanhui Chen, and Haiming Sun. 2012. Cue-induced craving and physiological reactions in recently and long-abstinent heroin-dependent patients. *Addictive behaviors* 37, 4 (2012), 393–398.